

UNIVERSITY OF MALAYA



WIX 1003 COMPUTER SYSTEM AND ORGANIZATION

SEMESTER 1, 2025/2026

LAB ASSIGNMENT REPORT

TRAFFIC LIGHT CONTROL SYSTEM

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Occurence: 10

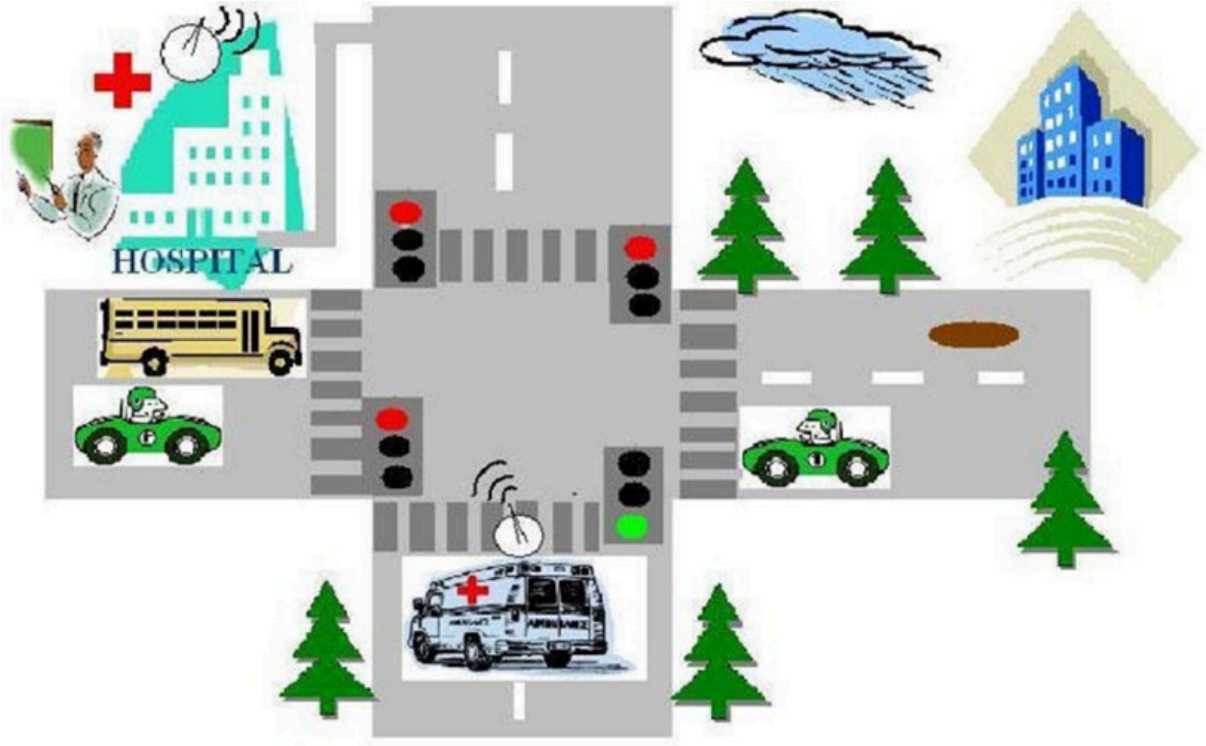
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1.0 Introduction



Our team has been assigned the task of developing a prototype for a new computer system capable of controlling traffic at a cross-junction. Specifically, we are required to produce a working demo of a traffic light control system for the junction and traffic flow illustrated in the figure below.

As a minimal requirement for this system, a microprocessor or microcontroller must be utilized. The system is designed to provide an equal opportunity for all users to exit and cross the junction safely. Each direction of the junction is regulated by a tri-color traffic light indicator (green, yellow, and red). The controller system manages these indicators to grant permission for traffic movement without causing accidents or havoc.

The signaling logic for the indicators is defined as follows:

- Green: Permission to go
- Yellow: Prepare to stop
- Red: Stop and wait

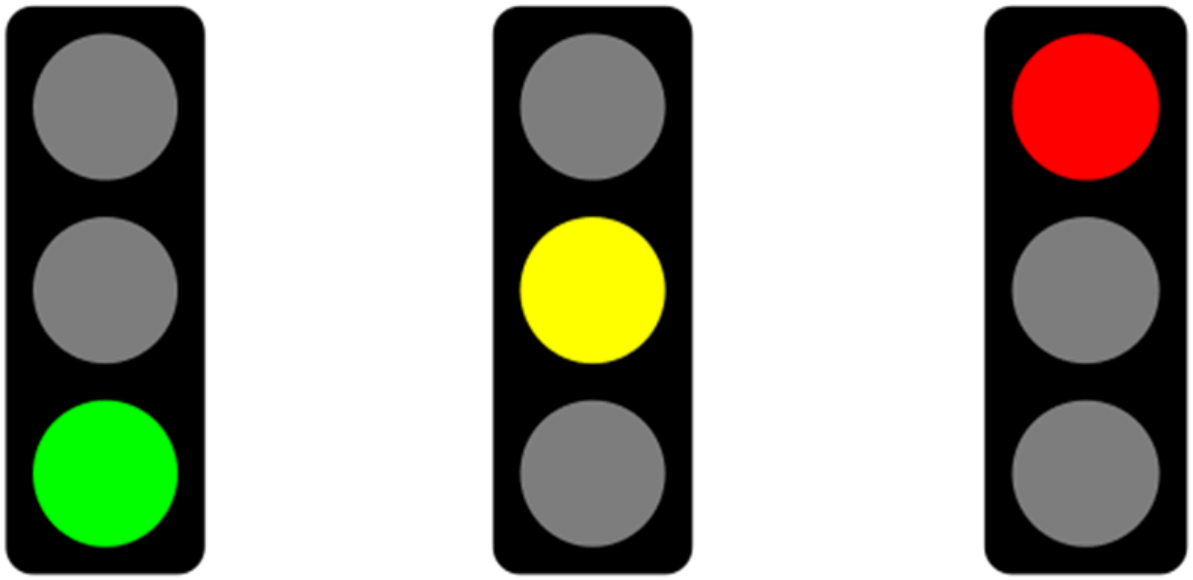
Additionally, a countdown indicator has been suggested to inform users of the remaining wait time before permission to proceed is granted, further enhancing the system's efficiency and user awareness.

2.0 What is Traffic Light

Traffic lights, also known as traffic signals, serve as crucial devices to regulate the movement of vehicles at road intersections and crossings. It is designed to ensure an orderly flow of traffic, provide an opportunity for pedestrians or vehicles to cross an intersection and help reduce the number of conflicts between vehicles entering intersections from different directions. These lights are equipped with different colors to convey specific instructions to drivers. The sequence of color changes follows a consistent order.

Three color light

- When the green light is on, it tells the driver to continue driving.
- When the yellow light is on, it serves as a warning, indicating that the driver should prepare to stop when safe to do so, as the light is about to transition to red.
- When the red light is on, it tells drivers to come to a complete stop.



Green : Permission to go Yellow : Prepare to stop Red : Stop and wait

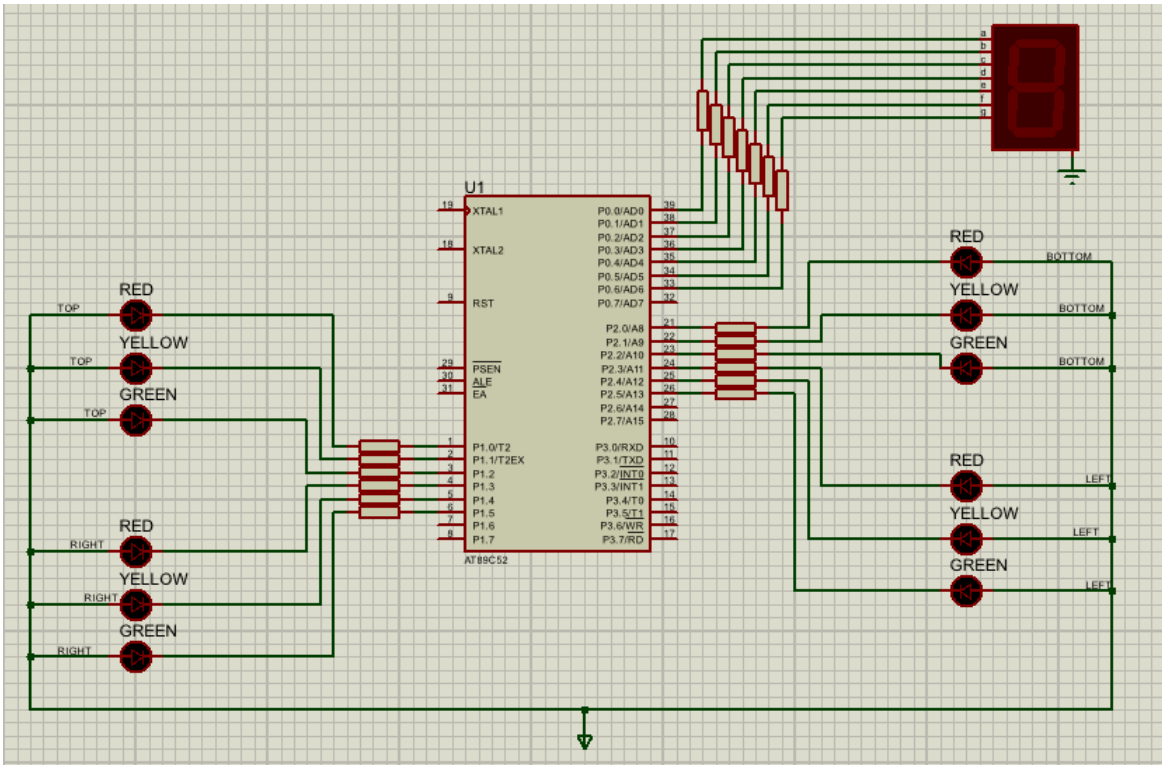
Countdown Lights

A traffic light countdown timer is a digital clock designed to display the remaining time of the current traffic light signal, whether it is red, yellow, or green. It is important as drivers could have preparation time to prepare to stop or proceed driving. This real-time information enhances driver readiness and more controlled flow of traffic at intersections.



Traffic countdown timer

3.0 Component Connection Diagram



4.0 Complete Code of Our System

```
;------------------------------------------------------------
; 4-Way Traffic Light (Active HIGH / Common Cathode)
; Chip: AT89S52
; Display Logic: 1 = ON, 0 = OFF
;------------------------------------------------------------

ORG 0000H
    LJMP MAIN

;------------------------------------------------------------
; MAIN LOOP
; Traffic Light Logic Active Low
; Display Logic Active High
;------------------------------------------------------------
MAIN:
    ;--- LANE 1 GREEN SEQUENCE ---
    MOV P1, #0DEH
    MOV P2, #0DBH
    ACALL COUNT_GREEN

    ;--- LANE 1 YELLOW SEQUENCE ---
    MOV P1, #0DDH
    ACALL COUNT_YELLOW

    ;--- LANE 2 GREEN SEQUENCE ---
    MOV P1, #0F3H
    MOV P2, #0DBH
    ACALL COUNT_GREEN

    ;--- LANE 2 YELLOW SEQUENCE ---
    MOV P1, #0EBH
    ACALL COUNT_YELLOW

    ;--- LANE 3 GREEN SEQUENCE ---
    MOV P1, #0DBH
    MOV P2, #0DEH
    ACALL COUNT_GREEN

    ;--- LANE 3 YELLOW SEQUENCE ---
    MOV P2, #0DDH
    ACALL COUNT_YELLOW

    ;--- LANE 4 GREEN SEQUENCE ---
    MOV P1, #0DBH
    MOV P2, #0F3H
    ACALL COUNT_GREEN

    ;--- LANE 4 YELLOW SEQUENCE ---
    MOV P2, #0EBH
    ACALL COUNT_YELLOW

    LJMP MAIN

;------------------------------------------------------------
; SUBROUTINE: Count 9 down to 3
;------------------------------------------------------------
COUNT_GREEN:
    MOV R0, #9
G_LOOP:
    MOV A, R0
    ACALL DISPLAY_NUM
    ACALL DELAY_1SEC
    DEC R0
    CJNE R0, #3, G_CONT
    RET
G_CONT:
    SJMP G_LOOP

;------------------------------------------------------------
; SUBROUTINE: Count 2 down to 0
;------------------------------------------------------------
COUNT_YELLOW:
```

```
    MOV A, R0
    ACALL DISPLAY_NUM
    ACALL DELAY_1SEC
    DEC R0
    CJNE R0, #0FFH, Y_CONT
    RET
Y_CONT:
    SJMP Y_LOOP

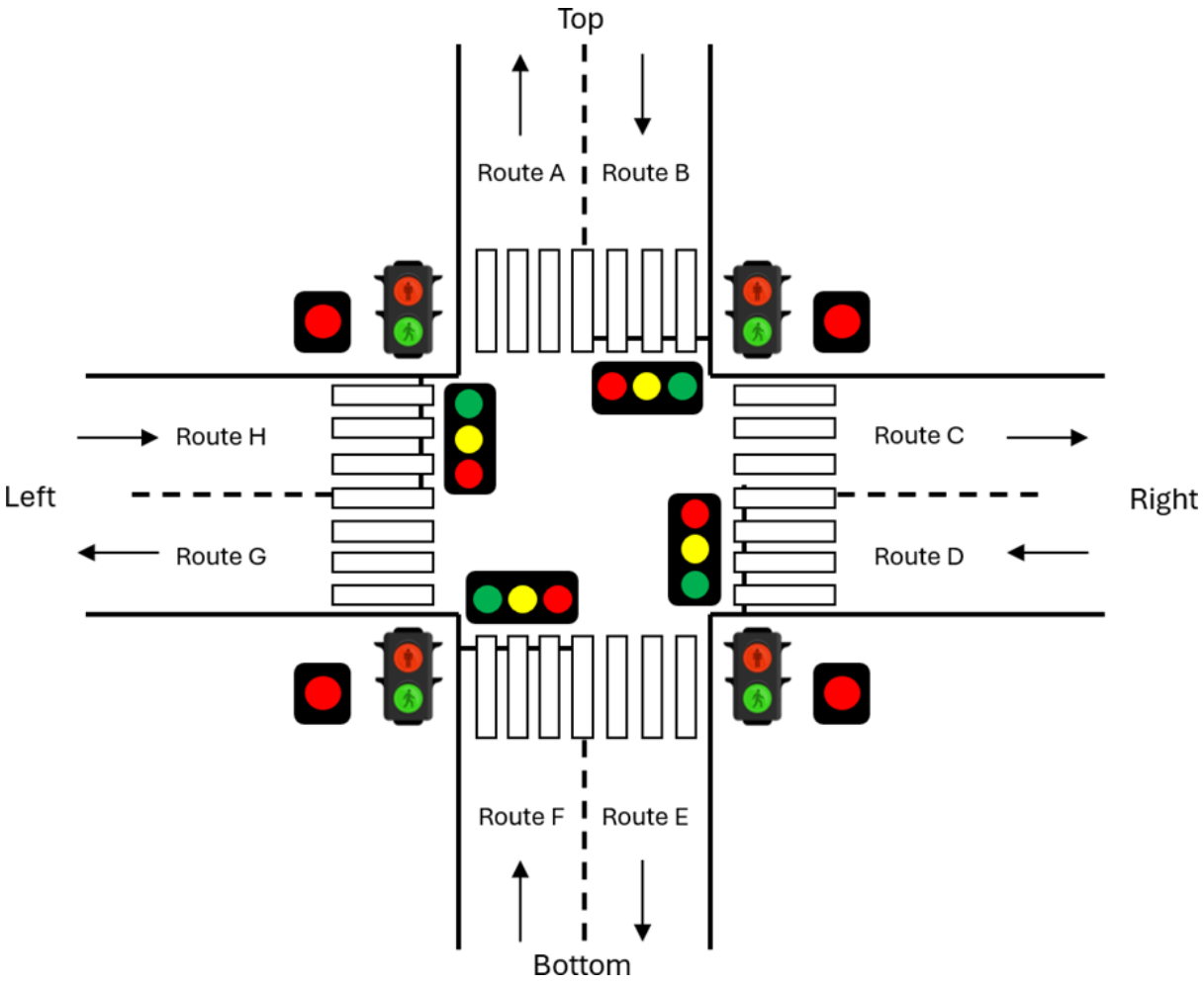
;-------------------------------------------------------
; SUBROUTINE: Display Number
;-------------------------------------------------------
DISPLAY_NUM:
    MOV DPTR, #NUM_TABLE
    MOVC A, @A+DPTR
    MOV P0, A               ; Output to Port 0
    RET

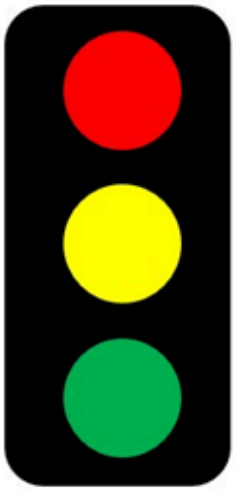
;-------------------------------------------------------
; LOOKUP TABLE: INVERTED (Active High / Common Cathode)
; 1 means segment ON, 0 means segment OFF
;-------------------------------------------------------
NUM_TABLE:
    DB 3FH  ; 0 (was C0)
    DB 5BH  ; 2 (was A4)
    DB 4FH  ; 3 (was B0)
    DB 66H  ; 4 (was 99)
    DB 6DH  ; 5 (was 92)
    DB 7DH  ; 6 (was 82)
    DB 07H  ; 7 (was F8)
    DB 7FH  ; 8 (was 80)
    DB 6FH  ; 9 (was 90)

;-------------------------------------------------------
; SUBROUTINE: 1 Second Delay
;-------------------------------------------------------
DELAY_1SEC:
    MOV R3, #10
D1: MOV R4, #200
D2: MOV R5, #250
D3: DJNZ R5, D3
    DJNZ R4, D2
    DJNZ R3, D1
    RET

END
```

5.0 Explaination of System Operation





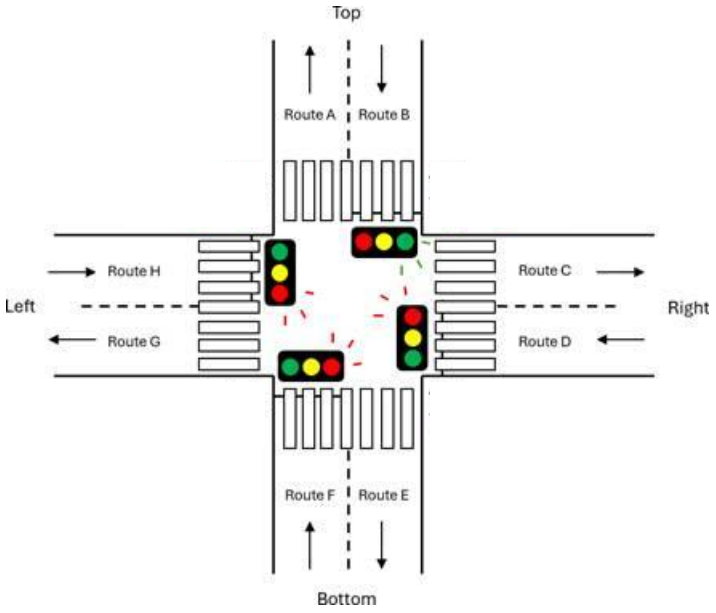
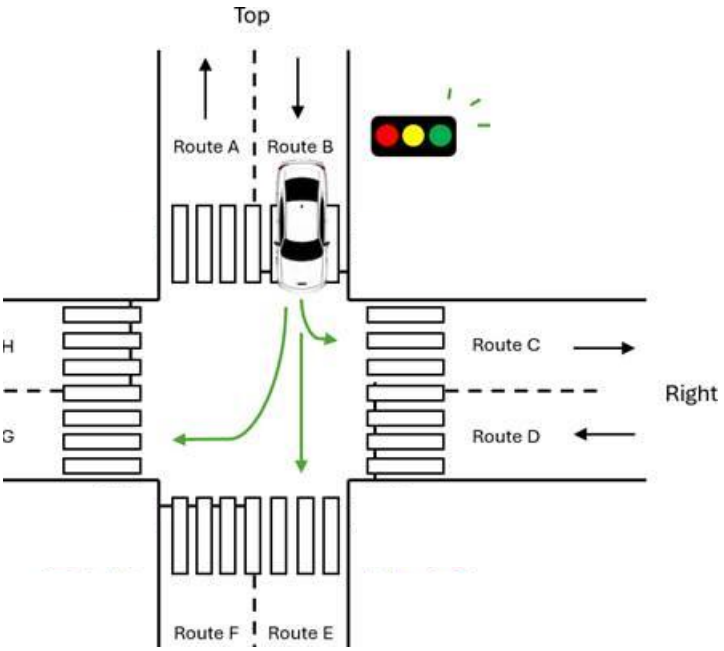
Red: Stop and wait

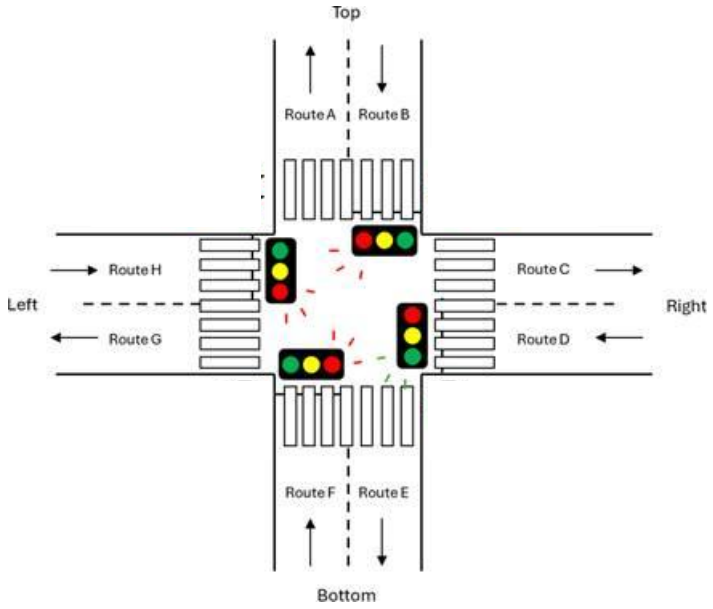
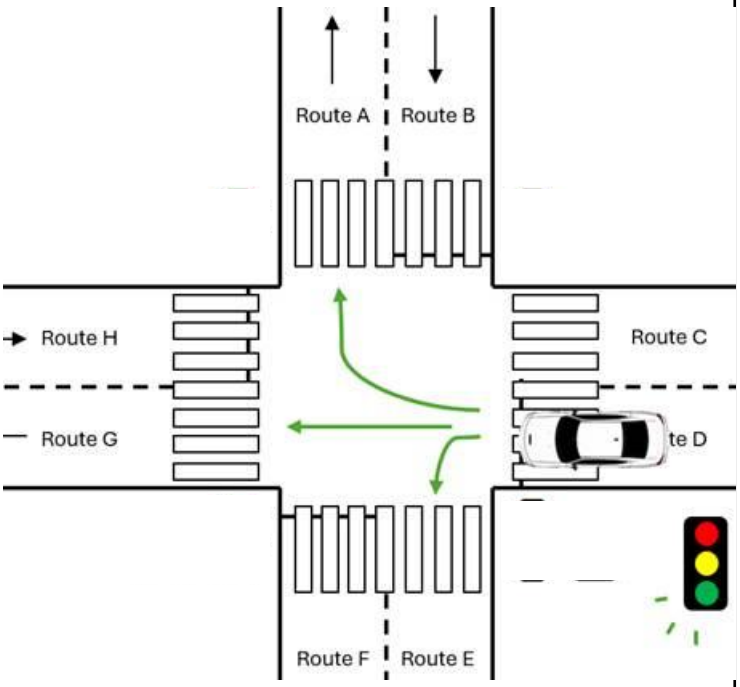
Yellow: Prepare to stop

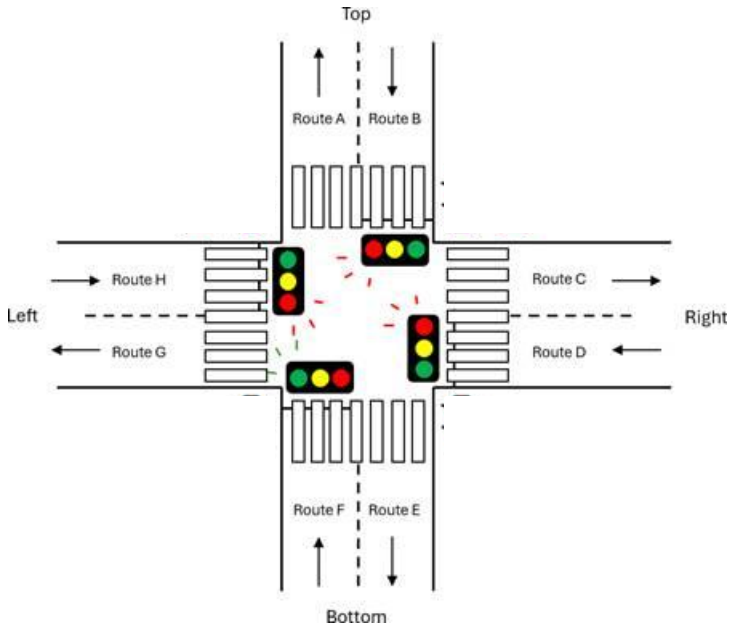
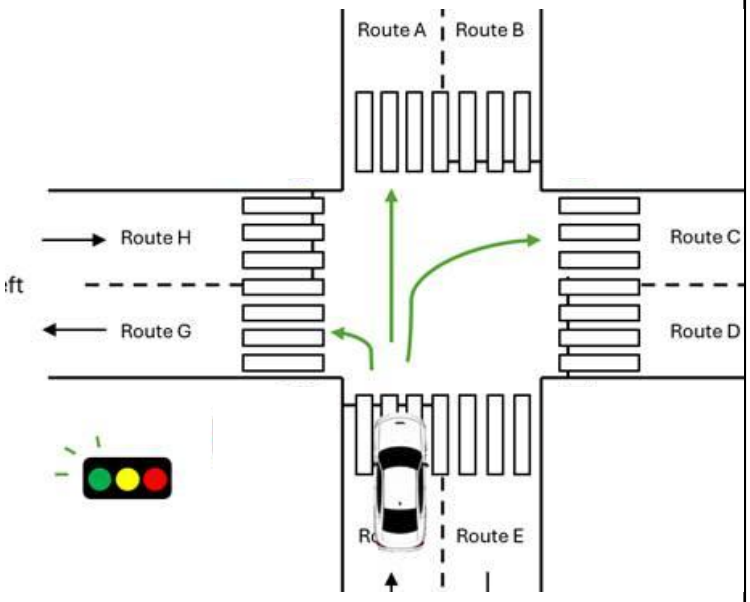
Green: Permission to go

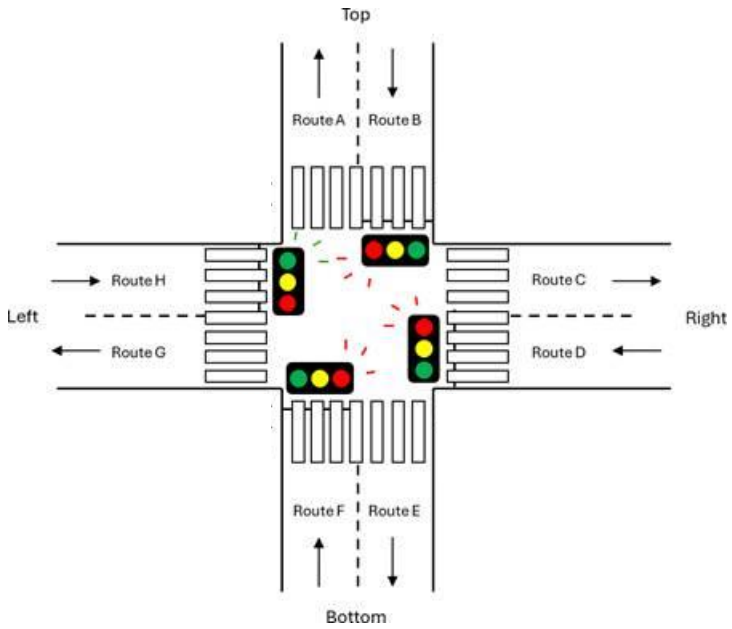
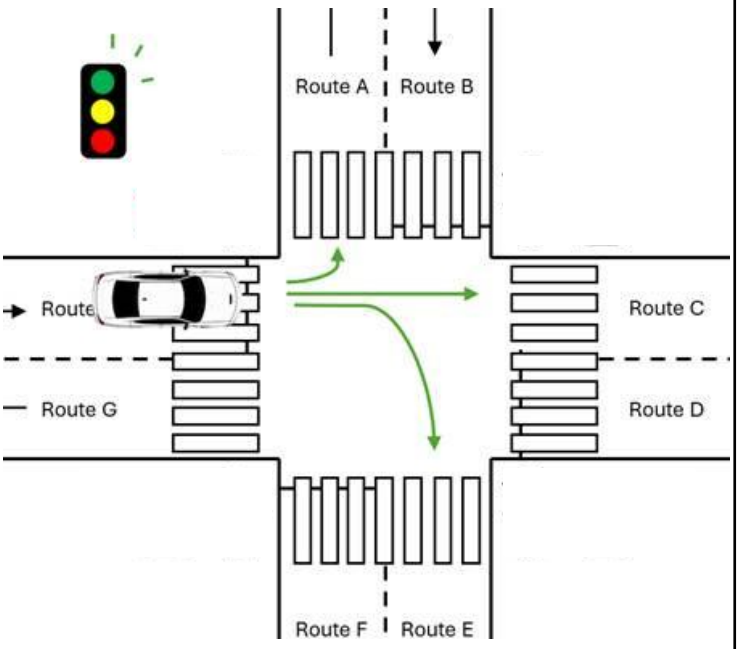
Traffic Light

Traffic Light:

Stage 1	Explanation
	TOP's traffic lights will turn green while others will remain red for 10 seconds. After 9 seconds, the TOP's traffic light will delay for 1 second then turn yellow for 3 seconds and delay another 1 second then turn red. When the TOP's traffic light turns red, the system moves to STAGE 2.
	**<u>Flows of traffic</u>** When lights is on green: Vehicles from route B are allowed to enter route C, E or G. Other vehicles must stop at each route respectively. When lights is on yellow: Vehicles from route B must prepare to stop before lights turn to red. Vehicles from route D ready to move. When lights is on red: Vehicles from route B must stop.

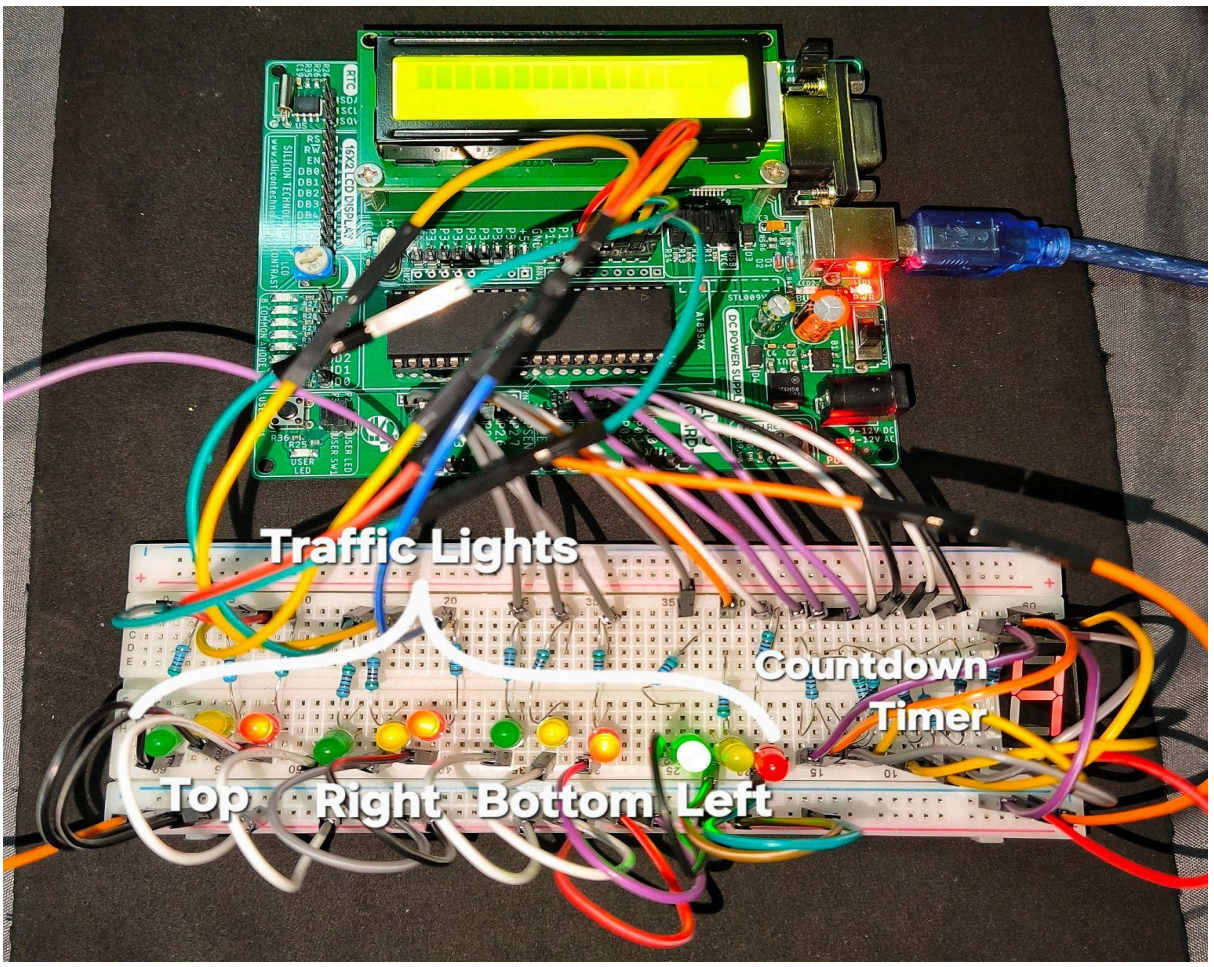
Stage 2	Explanation
	RIGHT's traffic lights will turn green while others will remain red for 10 seconds. After 9 seconds, the RIGHT's traffic light will delay for 1 second then turn yellow for 3 seconds and delay another 1 second then turn red. When the RIGHT's traffic light turns red, the system moves to STAGE 3.
	**<u>Flows of traffic</u>** When lights is on green: Vehicles from route D are allowed to enter route E, G or A. Other vehicles must stop at each route respectively. When lights is on yellow: Vehicles from route D must prepare to stop before lights turn to red. Vehicles from route F ready to move. When lights is on red: Vehicles from route D must stop.

Stage 3	Explanation
	BOTTOM's traffic lights will turn green while others will remain red for 10 seconds. After 9 seconds, the BOTTOM's traffic light will delay for 1 second then turn yellow for 3 seconds and delay another 1 second then turn red. When the BOTTOM's traffic light turns red, the system moves to STAGE 4.
	**<u>Flows of traffic</u>** When lights is on green: Vehicles from route F are allowed to enter route G, A or C. Other vehicles must stop at each route respectively. When lights is on yellow: Vehicles from route F must prepare to stop before lights turn to red. Vehicles from route H ready to move. When lights is on red: Vehicles from route F must stop.

Stage 4	Explanation
	LEFT's traffic lights will turn green while others will remain red for 10 seconds. After 9 seconds, the LEFT's traffic light will delay for 1 second then turn yellow for 3 seconds and delay another 1 second then turn red. When the LEFT's traffic light turns red, the system moves to STAGE 1.
	**<u>Flows of traffic</u>** When lights is on green: Vehicles from route H are allowed to enter route A, C or E. Other vehicles must stop at each route respectively. When lights is on yellow: Vehicles from route H must prepare to stop before lights turn to red. Vehicles from route B ready to move. When lights is on red: Vehicles from route H must stop.

6.0 Circuit Design

Image of Hardware Implementation



Video and Source Code

https://drive.google.com/drive/folders/1fi2IqUmnzAwSR2wIs5Xi8M-pPzeIhmy7

7.0 Design Considerations

7.1 Optimal Signal Placement and Visibility

Four tri-color traffic lights are positioned at each corner of the junction at an appropriate height. This placement is crucial for optimizing the intersection’s efficiency, as it ensures that all road users, regardless of their position in the queue, have an unobstructed view of the signals. By maximizing visibility, the system reduces the likelihood of confusion, promotes smoother traffic flow, and minimizes the risk of accidents.

7.2 User-Centric Safety Features

Each traffic light is equipped with an integrated countdown timer. This feature provides drivers with a clear visual indication of the remaining time before a signal change. From a safety perspective, this reduces "dilemma zone" incidents and helps prevent rear-end collisions, as drivers can better anticipate when they need to decelerate for a red light or prepare to move for a green light.

7.3 Fairness and Traffic Flow Logic

The system is programmed to provide an equal distribution of time for all directions, ensuring fairness and preventing long congestion build-ups on any single side. The timing logic is defined as follows:

- Green Light: 9 seconds (active movement).
- Yellow Light: 3 seconds (transition/caution).
- Red Light: Calculated based on the opposing junction's cycle to ensure no "havoc" or conflicting movements.

7.4 Precision Timing via MCU8051

For the working demo, the MCU8051 microcontroller is utilized to handle the system’s timing requirements. To achieve a precise one-second delay for the countdown, the system’s assembly logic calculates the necessary machine cycles based on the oscillator frequency. This ensures that the countdown and light transitions remain synchronized and accurate to real-time standards.

8.0 System Limitation

Limitation	Technical Explanation
Non-Realistic Timing Intervals	In this prototype, the Green light is programmed for 9 seconds and Yellow for 3 seconds to facilitate a quick demonstration. In real-world urban intersections, these durations are much longer and vary based on road size and speed limits.
Sequential Logic vs. Adaptive Control	The current Assembly code follows a strict sequential loop (MAIN). It lacks input from vehicle detection sensors (such as inductive loops or IR sensors). Consequently, the system cannot skip empty lanes or extend green lights for heavy traffic, leading to potential congestion.
Lack of Surveillance Integration	The system does not incorporate traffic cameras or logging modules. In professional systems, these are essential for monitoring traffic violations (red-light jumping) and providing data for city planning.
Hardware-Dependent Timing	The 1-second delay is generated using nested register loops (R3, R4, R5). While effective for a demo, this timing can fluctuate if the crystal oscillator frequency changes, unlike a real-world system that would use a dedicated Real-Time Clock (RTC).

9.0 Conclusion

We have successfully completed the assignment by fulfilling all the fundamental requirements for a 4-way traffic light control prototype. The system effectively manages traffic flow across a cross-junction using an AT89S52 microcontroller.

In addition to the basic tri-color signaling logic, we implemented a countdown indicator which is a 7-segment display (controlled via Port 0) that informs users of the remaining time, reducing driver anxiety and improving safety.

While the project has limitations regarding sensor integration and timing scale, it successfully demonstrates the core principles of Computer Systems & Organization and the practical application of Assembly language in embedded systems.